

[50 PowerShell Commands for SOC L1 Analyst (With Use Cases)]

[Core System & Log Commands]

1. Get-Process

```
Get-Process
```

Use: Show running processes

SOC Use Case: Check suspicious processes like malware or crypto miners.

2. Get-Service

```
Get-Service
```

Use: Show Windows services

SOC Use Case: Find suspicious or disabled security services.

3. Get-EventLog

```
Get-EventLog -LogName Security
```

Use: Read Windows logs

SOC Use Case: Investigate login attempts and attacks.

4. Get-WinEvent

```
Get-WinEvent -LogName Security
```

Use: Advanced event log search

SOC Use Case: Faster log hunting and filtering.

5. Get-LocalUser

```
Get-LocalUser
```

Use: List local users

SOC Use Case: Detect unknown accounts created by attackers.

6. Get-LocalGroupMember

```
Get-LocalGroupMember Administrators
```

Use: Show group members

SOC Use Case: Find unauthorized admin users.

7. whoami

```
whoami
```

Use: Show current user

SOC Use Case: Verify current account during investigation.

8. hostname

```
hostname
```

Use: Show system name

SOC Use Case: Identify infected machine.

9. ipconfig

```
ipconfig /all
```

Use: Show IP configuration

SOC Use Case: Check suspicious DNS or IP settings.

10. Get-NetIPAddress

```
Get-NetIPAddress
```

Use: Display IP addresses

SOC Use Case: Investigate network configuration.

[Network Investigation Commands]

11. netstat -ano

```
netstat -ano
```

Use: Show active connections

SOC Use Case: Detect malware communicating with attacker IPs.

12. Get-NetTCPConnection

```
Get-NetTCPConnection
```

Use: View TCP connections

SOC Use Case: Analyze suspicious ports.

13. Test-NetConnection

```
Test-NetConnection google.com
```

Use: Test connectivity

SOC Use Case: Verify outbound network access.

14. Resolve-DnsName

```
Resolve-DnsName google.com
```

Use: DNS lookup

SOC Use Case: Investigate malicious domains.

15. nslookup

```
nslookup example.com
```

Use: Query DNS records

SOC Use Case: Threat intelligence investigation.

16. tracert

```
tracert google.com
```

Use: Trace route

SOC Use Case: Analyze suspicious traffic paths.

17. ping

```
ping 8.8.8.8
```

Use: Check connectivity

SOC Use Case: Verify host communication.

18. arp -a

```
arp -a
```

Use: Show ARP table

SOC Use Case: Detect suspicious local devices.

19. route print

```
route print
```

Use: Show routing table

SOC Use Case: Identify malicious routes.

20. Get-NetAdapter

```
Get-NetAdapter
```

Use: Show network adapters

SOC Use Case: Check unknown virtual adapters created by malware.

[File Investigation Commands]

21. Get-ChildItem

```
Get-ChildItem
```

Use: List files/folders

SOC Use Case: Investigate suspicious directories.

22. Get-Content

```
Get-Content log.txt
```

Use: Read file contents

SOC Use Case: Analyze logs and scripts.

23. Select-String

```
Select-String "error" log.txt
```

Use: Search inside files

SOC Use Case: Hunt indicators of compromise.

24. Copy-Item

```
Copy-Item test.txt C:\Temp
```

Use: Copy files

SOC Use Case: Backup suspicious files for analysis.

25. Move-Item

```
Move-Item malware.exe C:\Quarantine
```

Use: Move files

SOC Use Case: Quarantine suspicious files.

26. Remove-Item

```
Remove-Item temp.ps1
```

Use: Delete files

SOC Use Case: Remove malicious scripts.

27. Rename-Item

```
Rename-Item file.txt newfile.txt
```

Use: Rename files

SOC Use Case: Organize investigation evidence.

28. Get-FileHash

```
Get-FileHash malware.exe
```

Use: Calculate file hash

SOC Use Case: Compare hashes with threat intel.

29. Compress-Archive

```
Compress-Archive logs C:\logs.zip
```

Use: Compress files

SOC Use Case: Send evidence securely.

30. Expand-Archive

```
Expand-Archive logs.zip
```

Use: Extract ZIP files

SOC Use Case: Extract collected logs.

[Process & System Monitoring]

31. tasklist

```
tasklist
```

Use: Show running tasks

SOC Use Case: Identify suspicious processes.

32. taskkill

```
taskkill /PID 1234 /F
```

Use: Kill process

SOC Use Case: Stop malicious activity.

33. Get-CimInstance

```
Get-CimInstance Win32_OperatingSystem
```

Use: System information

SOC Use Case: Collect forensic system details.

34. Get-ComputerInfo

```
Get-ComputerInfo
```

Use: Full system info

SOC Use Case: Asset identification.

35. systeminfo

```
systeminfo
```

Use: System details

SOC Use Case: Gather machine information quickly.

36. Get-HotFix

```
Get-HotFix
```

Use: Show installed patches

SOC Use Case: Check missing security updates.

37. Get-ScheduledTask

```
Get-ScheduledTask
```

Use: View scheduled tasks

SOC Use Case: Detect persistence mechanisms.

38. schtasks

```
schtasks /query
```

Use: Manage scheduled tasks

SOC Use Case: Investigate malicious automation.

39. Get-ExecutionPolicy

```
Get-ExecutionPolicy
```

Use: Show PowerShell policy

SOC Use Case: Detect insecure script settings.

40. Set-ExecutionPolicy

```
Set-ExecutionPolicy RemoteSigned
```

Use: Change execution policy

SOC Use Case: Secure PowerShell environment.

[Security Investigation Commands]

41. Get-MpComputerStatus

```
Get-MpComputerStatus
```

Use: Windows Defender status

SOC Use Case: Verify antivirus health.

42. Get-MpThreat

```
Get-MpThreat
```

Use: Show detected threats

SOC Use Case: Investigate malware alerts.

43. Get-History

```
Get-History
```

Use: Show PowerShell history

SOC Use Case: Review attacker commands.

44. Clear-History

```
Clear-History
```

Use: Clear command history

SOC Use Case: Security cleanup after testing.

45. Get-Clipboard

```
Get-Clipboard
```

Use: Read clipboard content

SOC Use Case: Detect copied credentials or malware commands.

46. Start-Transcript

```
Start-Transcript
```

Use: Record PowerShell session

SOC Use Case: Maintain investigation logs.

47. Stop-Transcript

```
Stop-Transcript
```

Use: Stop recording session

SOC Use Case: Save investigation evidence.

48. Get-Command

```
Get-Command
```

Use: List available commands

SOC Use Case: Discover useful security commands.

49. Get-Help

```
Get-Help Get-Process
```

Use: Command help

SOC Use Case: Learn command usage quickly.

50. Invoke-History

```
Invoke-History 3
```

Use: Re-run command from history

SOC Use Case: Repeat investigation commands efficiently.

[Most Important Commands for SOC L1]

If you are beginner SOC analyst, focus first on these:

- 1. `Get-EventLog`
- 2. `Get-WinEvent`
- 3. `Get-Process`
- 4. `netstat -ano`
- 5. `Get-NetTCPConnection`
- 6. `Get-FileHash`
- 7. `Get-ScheduledTask`
- 8. `Get-LocalUser`
- 9. `tasklist`
- 10. `Get-MpThreat`